

JAMES MITCHELL

Bristol, United Kingdom

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 14 years of hands-on experience building and deploying production ML systems across fintech, healthcare, and e-commerce sectors. Proven track record of leading cross-functional teams to deliver data-driven solutions that directly impact business metrics. Skilled at translating complex statistical concepts into practical implementations while mentoring junior engineers and establishing ML best practices within organizations.

WORK EXPERIENCE

Staff Machine Learning Engineer

DataFlow Systems Ltd

www.dataflowsystems.io

London, United Kingdom

March 2021 - Present

Architected and deployed a real-time fraud detection system handling 50 million daily transactions, reducing false positives by 34% through ensemble modeling techniques and improving customer experience across 2.3 million active users.

Led a team of 5 ML engineers in building a recommendation engine that increased product cross-sell revenue by 18% within the first six months of production deployment.

Established MLOps infrastructure using Kubernetes and Airflow, reducing model deployment time from 3 weeks to 2 days and enabling automated retraining pipelines for 40+ models.

Mentored 8 junior data scientists and engineers, conducting weekly technical reviews and creating internal documentation that became the foundation for the company's ML engineering standards.

Senior Machine Learning Engineer

HealthTech Innovations plc

www.healthtechinnovations.com

Cambridge, United Kingdom

July 2017 - February 2021

Developed and validated a machine learning model for early disease detection that achieved 96% sensitivity and 92% specificity on holdout test sets, directly contributing to regulatory approval for clinical use.

Implemented feature engineering pipelines processing 500GB of patient data weekly, reducing model training time by 60% through careful data preprocessing and dimensionality reduction.

Collaborated with product and clinical teams to translate business requirements into technical ML specifications, ensuring models addressed actual clinical needs rather than optimizing for metrics alone.

Built comprehensive evaluation frameworks and conducted extensive ablation studies that influenced architectural decisions for 3 major products in the company's portfolio.

Machine Learning Engineer

CloudVision Analytics Ltd

www.cloudvision.co.net

Bristol, United Kingdom

September 2010 - June 2017

Developed computer vision models for automated quality inspection in manufacturing, reducing defect detection time from 8 hours to 15 minutes per production batch.

Created data annotation workflows and quality assurance processes that improved training data consistency, resulting in 8% improvement in model accuracy across the computer vision product line.

Optimized model inference time through quantization and pruning techniques, enabling deployment on edge devices and reducing cloud infrastructure costs by 42%.

Designed and delivered training workshops on machine learning fundamentals to 120+ engineers across the organization, establishing a foundation for ML adoption across different business units.

EDUCATION

Doctor of Philosophy in Computer Science

University of Manchester

Manchester, United Kingdom

2008 - 2010

Thesis: Deep Learning Approaches for Sequential Data Analysis

Bachelor of Science in Mathematics and Statistics

University of Bristol

Bristol, United Kingdom

2004 - 2008

TECHNICAL SKILLS

Machine Learning: TensorFlow, PyTorch, scikit-learn, XGBoost, LightGBM, neural networks, ensemble methods, deep learning, reinforcement learning, recommendation systems

Data Processing: Python, Pandas, NumPy, Spark, SQL, Apache Airflow, feature engineering, data validation

ML Operations: Kubernetes, Docker, model deployment, CI/CD pipelines, experiment tracking, model monitoring, A/B testing

Computer Vision: CNN architectures, image classification, object detection, segmentation, transfer learning

Natural Language Processing: transformer models, BERT, text classification, embedding techniques

Cloud Platforms: AWS SageMaker, Google Cloud ML Engine, Azure ML Studio

Statistical Methods: Bayesian inference, hypothesis testing, statistical modeling, causal inference